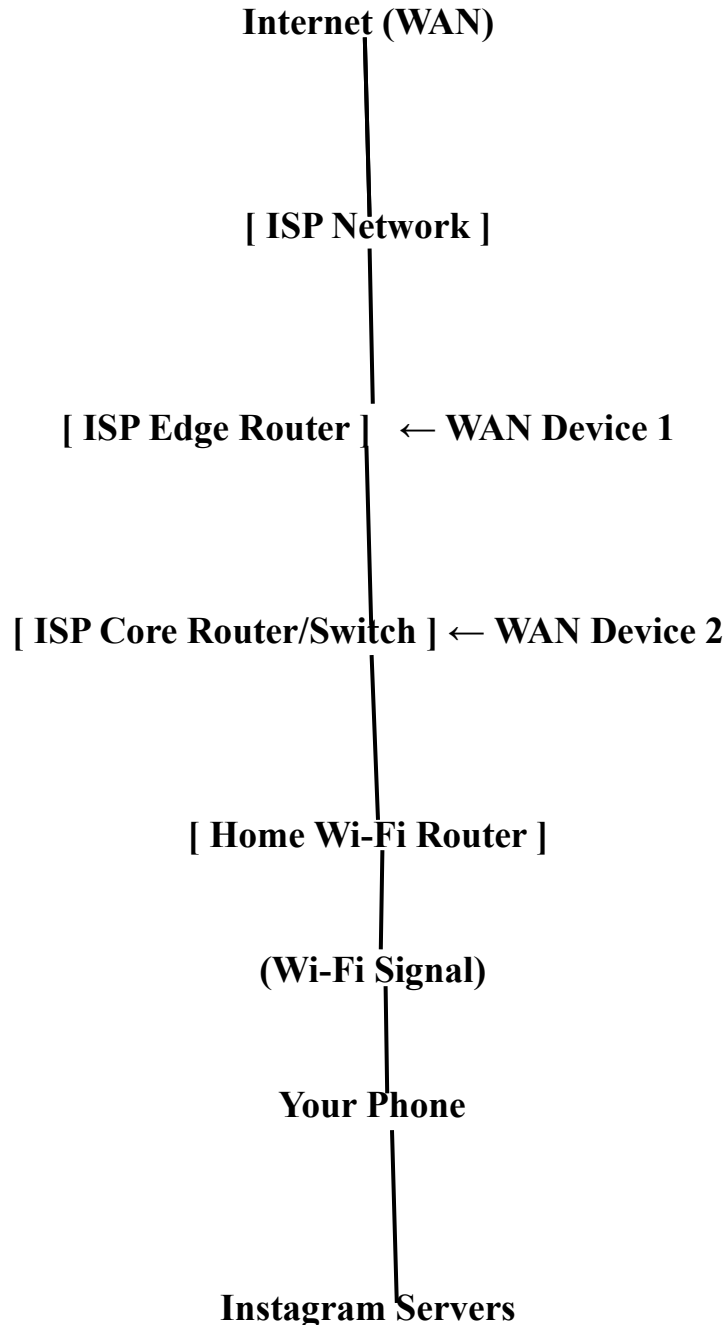


WIDE AREA NETWORK

1. Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.



2. Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

WAN Technology	Speed	Cost	Real-Life Use
MPLS (Multiprotocol Label Switching)	High	High	Used by large companies like Zomato offices, for secure and reliable communication.
Leased Line	Very High (Dedicated Connection)	Very High	Used by IRCTC ticketing systems, data centers, where an uninterrupted connection is essential.
Broadband Internet	Medium to High	Low	Used in homes, small businesses, schools, and cafés for everyday internet access.

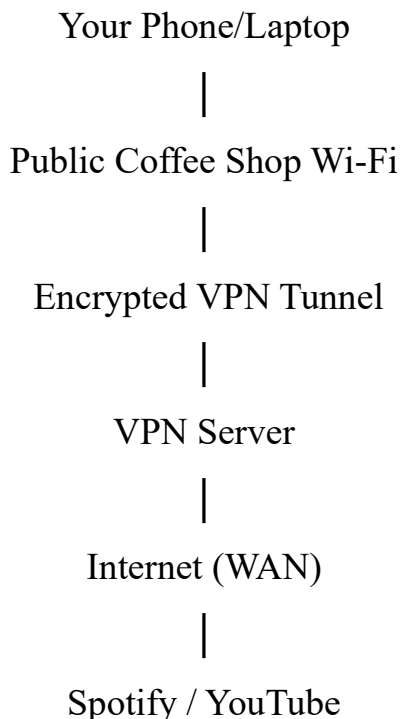
Comparison Summary

Feature	MPLS	Leased Line	Broadband Internet
Speed	High	Very High	Medium to High
Cost	High	Very High	Low
Reliability	High	Very High	Moderate

**3. Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.

Hint: Focus on how the VPN creates a private tunnel over the public WAN.**

- A VPN (Virtual Private Network) is a service that creates a secure, encrypted tunnel between your device and a VPN server over the public WAN (Internet).
- For example, when you connect to the public Wi-Fi at a coffee shop and open Spotify or YouTube, your data normally travels over the public Internet, where it could be intercepted by attackers. If you use a VPN, all your internet traffic is encrypted before it leaves your device.
- The VPN securely sends your data through a private tunnel across the public WAN to the VPN server. The VPN server then connects to Spotify or YouTube on your behalf and sends the encrypted data back to you. This helps protect your passwords, personal information, and browsing activity from others on the same public Wi-Fi network.



4. Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

For Flipkart to connect its warehouses in Ahmedabad, Mumbai, and Delhi, the best WAN technology is MPLS (Multiprotocol Label Switching).

Reasons:

1. High Security: MPLS provides a secure and private network for transferring important business data, such as inventory, orders, and customer information.
2. Reliable and Fast Connection: MPLS offers low latency and stable performance, ensuring smooth communication and real-time synchronization between warehouses.

Conclusion

Recommended WAN Technology: MPLS

It is the best choice because it provides secure communication and reliable, high-speed connectivity between Flipkart's warehouses in different cities.